Chapter 7: UNIT ROOT TESTS

- We have seen that a random walk with drift and the determinist trend processes are non-stationary, and their mean increases linearly over time. However, the source of the non-stationarity is different. The first one have a stochastic trend (has a unit root), the second one do not have a unit root.
- Transitory versus Permanent shocks:
  - Deterministic trend model: shocks are transitory;
  - Random walk processes: shocks are permanent.
- How can we identify the source of non-stationarity?
  - Unit root test can allow us to identify which type of non-stationary process we have.

Unit Root Tests

Tests with null hypothesis of unit roots

- Dickey-Fuller (DF)
- Augmented Dickey-Fuller(ADF)
- Phillips-Perron (PP)
- Dickey-Fuller GLS (DF-GLS)

Test with null hypothesis of no-unit roots

- Kwiatkowski, Phillips, Schmidt and Shin Test (KPSS)

UNIT ROOT TESTS: AR(1)

- Let's start with a nesting model:

$$X_t = \beta_0 + \beta_1 X_{t-1} + \beta_2 t + e_t$$

Restriction on the nesting model			
Zero-mean stationary AR(1)	$\beta_0 = 0$	$\lvert\beta_1\rvert < 1$	$\beta_2 = 0$
Non-zero mean stationary AR(1)	$\beta_0 \neq 0$	$\lvert\beta_1\rvert < 1$	$\beta_2 = 0$
Random Walk (RW)	$\beta_0 = 0$	$\beta_1 = 1$	$\beta_2 = 0$
Random Walk with Drift (RWWD)	$\beta_0 \neq 0$	$\beta_1 = 1$	$\beta_2 = 0$
Deterministic Trend (DT)		$\lvert\beta_1\rvert < 1$	$\beta_2 \neq 0$

UNIT ROOT TESTS

- Consider the following process:

$$X_t = \beta X_{t-1} + e_t$$

- Suppose we want to test the null hypothesis that β is equal to zero. By assuming the errors are iid (0, σ^2), under the null, this equation can be estimated by OLS and we can use the t-stat to test for significance.
- However, if $\beta = 1$ under the null, then we have non-stationarity. We have seen that the variance of a such process increases as t increases. In this case, under the null of a unit root we cannot use classical statistical methods to perform significance test on β (the estimate value of β is biased to be below its true value).
- The autocorrelogram of a random walk and an AR(1) near unit root process are similar. By only check the ACF might leas us to falsely conclude that the data was generated from a stationary process.

Dickey-Fuller unit root test (command dfuller on STATA)

The Dickey-Fuller test the null hypothesis of β_1=1

$$X_t = \beta_0 + \beta_1 X_{t-1} + \beta_2 t + e_t$$

Or

$\gamma = 0$ in the following equation:

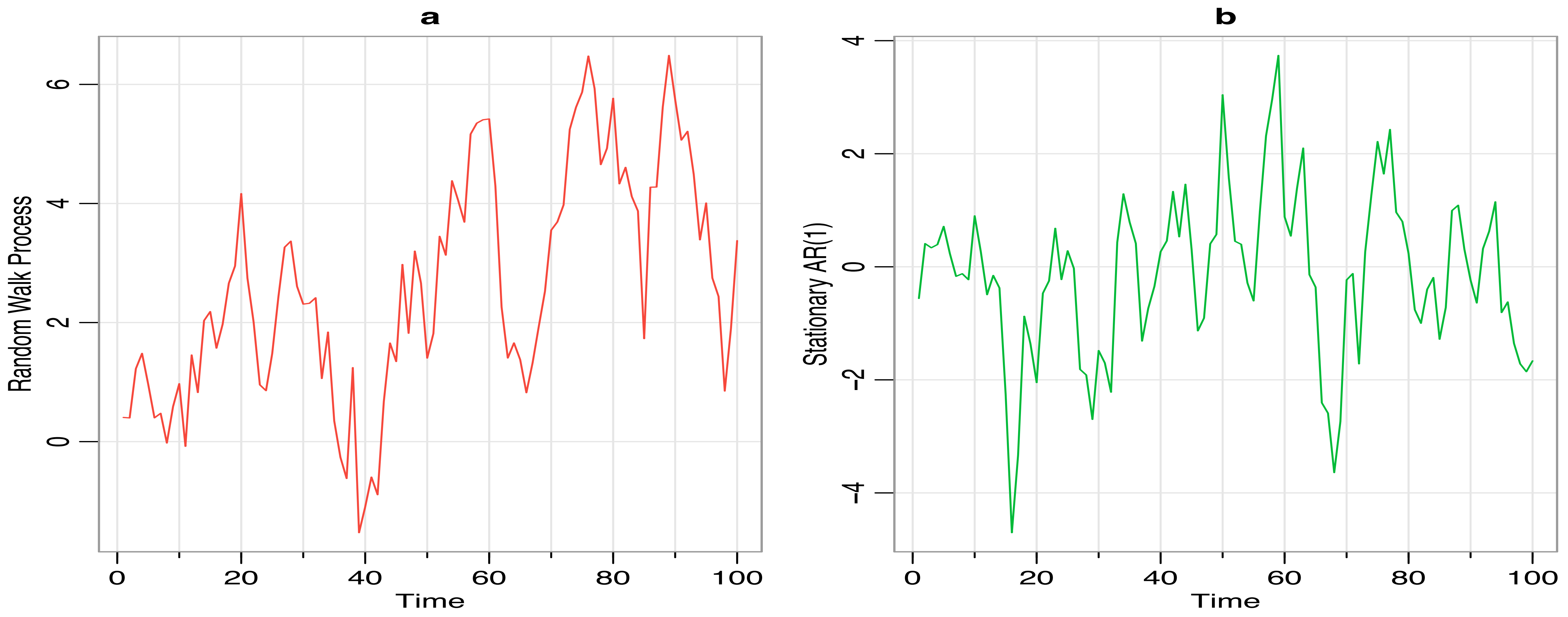
$$\Delta X_t = \gamma X_{t-1} + \beta_2 + e_t$$

Important: under the null hypothesis the distribution of β_1 does not follow a t-distribution (the distribution is function of a standard Brownian process (Wiener[1])). Critical values depend on the sample size and on the nested model (the inclusion of deterministic terms in the test regressions).

1- A Wiener process W(r) is a continuous-time stochastic process; $r \in [0,1]$.

- We estimate β by OLS, however, if the process has a unit root, then $T(\hat{\beta} - 1)$ converges in probability to $W(t) \sim N(0, t)$. It has a non-normal asymptotic distribution.

- The distribution depends on the nature of the deterministic components include (constant, trend). When including constant and trend, the 5% left-tail critical value is -3.41. The distribution is very skewed with a long tail on the left of $\beta = 1$.
- Critical values are tabulated by simulation techiques.



a
Random Walk Process
Time
0
2
4
6
0
20
40
60
80
100
b
Stationary AR(1)
Time
−4
−2
0
2
4
0
20
40
60
80
100

DF results: Random Walk Process without a drift (a)

```
dfuller X, noconstant regress

Dickey-Fuller test for unit root                   Number of obs   =        99

                         ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.306            -2.600            -1.950            -1.610

------------------------------------------------------------------------------
         D.X |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
           X |
         L1. |  -.0390038   .0298589    -1.31   0.195    -.0982579    .0202503
------------------------------------------------------------------------------
```

DF results: AR(1) Process (b) (without a mean)

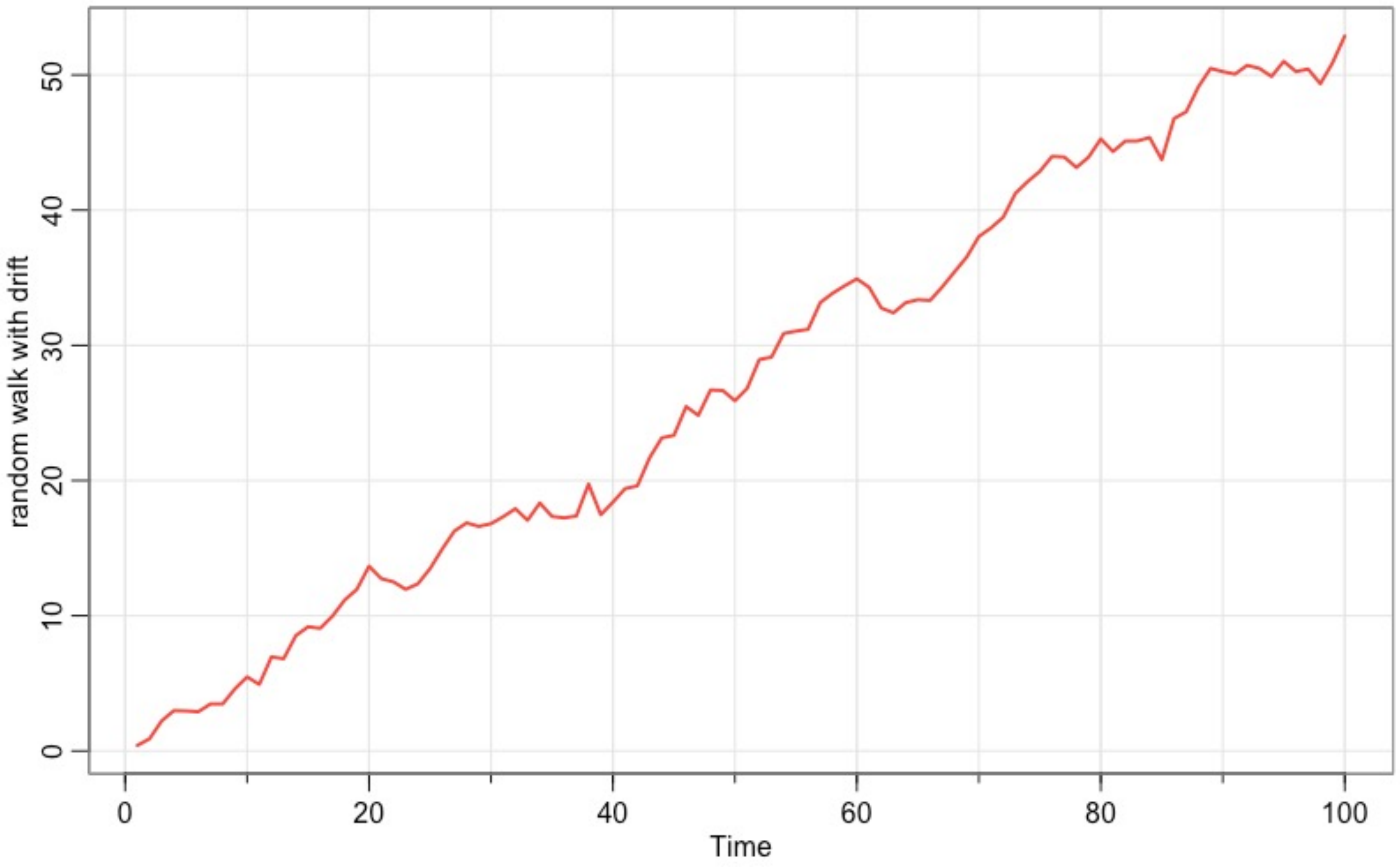
```
dfuller X, noconstant regress

Dickey-Fuller test for unit root                   Number of obs   =       99

                        ---------- Interpolated Dickey-Fuller ---------
              Test         1% Critical       5% Critical      10% Critical
            Statistic          Value             Value             Value
------------------------------------------------------------------------------
 Z(t)           -3.973           -2.600           -1.950           -1.610


------------------------------------------------------------------------------
        D.X |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
          X |
        L1. |  -.2832969   .0713082    -3.97   0.000    -.4248057    -.141788
------------------------------------------------------------------------------
```

Random walk with drift: $Y_t = 0.5 + Y_{t-1} + e_t$

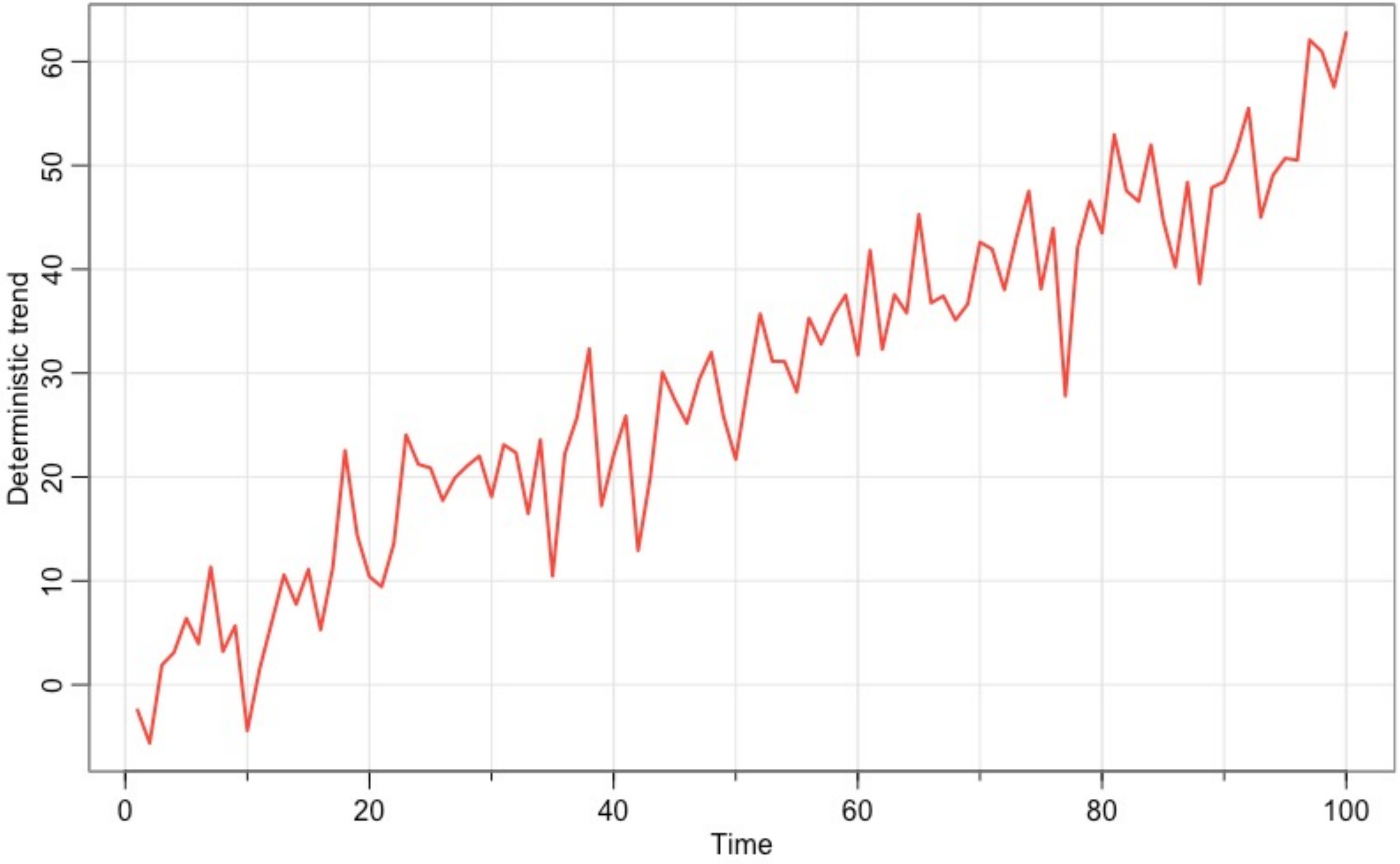


DF results: Random Walk Process with a drift

```
dfuller Y, regress trend
Dickey-Fuller test for unit root                 Number of obs   =       99

                 ---------- Interpolated Dickey-Fuller ---------
                  Test        1% Critical       5% Critical      10% Critical
               Statistic         Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -3.240           -4.042            -3.451            -3.151
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0768
------------------------------------------------------------------------------
        D.Y |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       Y L1.  |  -.2001429   .0617792    -3.24   0.002    -.3227737   -.0775122
     _trend |   .1071575   .0335155     3.20   0.002     .0406298    .1736852
      _cons |   .5990923   .1879078     3.19   0.002     .2260983    .9720862
------------------------------------------------------------------------------
```

Deterministic trend: $X_t = 1 + 0.10X_{t-1} + 0.5t + e_t$



DF results: AR(1) with a trend

```
dfuller X, regress trend
Dickey-Fuller test for unit root                Number of obs   =       99
                         ---------- Interpolated Dickey-Fuller ---------
              Test        1% Critical      5% Critical      10% Critical
           Statistic         Value            Value            Value
------------------------------------------------------------------------------
 Z(t)         -8.562          -4.042          -3.451          -3.151
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000


------------------------------------------------------------------------------
        D.X |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
      X  L1. |  -.8712877   .1017596    -8.56   0.000    -1.073279   -.6692966
     _trend |   .4705943   .0574803     8.19   0.000     .3564968    .5846919
      _cons |   2.207834   1.014744     2.18   0.032      .193583    4.222084
------------------------------------------------------------------------------
```

The ADF (Augmented) unit root test

- The original Dickey-Fuller test assumed that the error term was not serially correlated. However, this assumption is too strong (most of the times it is not valid).

- To account for autocorrelation in the error term we need an AR(p) instead of AR(1). The nested model becomes:

$$X_t = \alpha X_{t-1} + \beta_0 + \beta_2 t + \sum_{i=1}^{p-1} c_i \Delta X_{t-i} + e_t \quad \text{or}$$

$$\Delta X_t = \gamma X_{t-1} + \beta_0 + \beta_2 t + \sum_{i=1}^{k} c_i \Delta X_{t-i} + e_t$$

- Where: $\gamma = \left(\sum_{i=1}^{p} \beta_i - 1\right) = 0$

- How many lags (k) should we include? There are several different approaches.

Assuming we have an AR(p)

$$X_t = \beta_1 X_{t-1} + \beta_2 X_{t-2} + \cdots + \beta_p X_{t-p} + e_t$$

$$X_t = \sum_{i=1}^{p} \beta_i X_{t-i} + e_t$$

$$A(L)X_t = e_t$$

$$A(L) = 1 - \beta_1 L + \beta_2 L^2 + \cdots + \beta_p L^p$$

If the process has a unit root, then: $A(L) = (1 - L)A^*(L)$; $\quad A^*(L)$ has (p-1) roots outside the unit circle.

$$A(1) = 0 \Rightarrow 1 - \beta_1 + \beta_2 + \cdots + \beta_p = 0$$

$$\alpha = \sum_{i=1}^{p} \beta_i = 1$$

AR(p)

- By reparameterize the model AR(p) we will have:

$$X_t = \alpha X_{t-1} + \sum_{i=1}^{p-1} c_i \Delta X_{t-i} + e_t$$

Where : $c_i = -\sum_{j=i+1}^{p} \beta_i$

- Under the null hypothesis, we will test if $\alpha = 1$ against the alternative that $\alpha < 1$

Example: AR(2)

$$X_t = \beta_1 X_{t-1} + \beta_2 X_{t-2} + e_t$$

Add and substract $\beta_2 X_{t-1}$ to the right-hand side of the equation above:

$$X_t = \beta_1 X_{t-1} + \beta_2 X_{t-1} - \beta_2 X_{t-1} + \beta_2 X_{t-2} + e_t$$

$$X_t = (\beta_1 + \beta_2) X_{t-1} - \beta_2 \Delta X_{t-1} + e_t$$

Call $(\beta_1 + \beta_2) = \alpha$

$$X_t = \alpha X_{t-1} - \beta_2 \Delta X_{t-1} + e_t$$

ADF Unit Roots: AR(2)

```
dfuller Y, noconstant lags(1)regress

Augmented Dickey-Fuller test for unit root        Number of obs   =      2998
                   ---------- Interpolated Dickey-Fuller ---------
              Test         1% Critical       5% Critical      10% Critical
            Statistic          Value             Value             Value
------------------------------------------------------------------------------
 Z(t)          -11.785          -2.580          -1.950          -1.620


------------------------------------------------------------------------------
        D.Y |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       Y L1. |  -.1106199   .0093865   -11.78   0.000    -.1290246   -.0922152
         LD. |  -.1933389   .0179244   -10.79   0.000    -.2284843   -.1581935
------------------------------------------------------------------------------
```

Phillips-Perron unit root test (command pperron on STATA)

- The Phillips-Perron test is an alternative to the ADF test. Instead of adding lagged differences to compensate for serial correlation, it uses a non-parametric approach by correcting the standard errors for heteroskedasticity and autocorrelation (HAC): the Phillips-Perron test changes the test statistics instead of the regression.

- Unfortunately the Phillips-Perron test is known to have a severe size distortion (ADF also, but not as much) when the error has large negative MA root (the size distortion is toward over-rejecting the null hypothesis).

pperron Y, regress trend

Phillips-Perron test for unit root Number of obs = 99

Newey-West lags = 3

---------- Interpolated Dickey-Fuller ---------

	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(rho)	-22.332	-27.366	-20.682	-17.486
Z(t)	-3.427	-4.042	-3.451	-3.151

MacKinnon approximate p-value for Z(t) = 0.0478

Y	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
Y L1.	.7998571	.0617792	12.95	0.000	.6772263	.9224878
_trend	.1071575	.0335155	3.20	0.002	.0406298	.1736852
_cons	.5990923	.1879078	3.19	0.002	.2260983	.9720862

pperron D.Y, trend

Phillips-Perron test for unit root Number of obs = 98
Newey-West lags = 3

	---------- Interpolated Dickey-Fuller ---------			
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(rho)	-117.038	-27.332	-20.664	-17.472
Z(t)	-11.125	-4.044	-3.452	-3.151

MacKinnon approximate p-value for Z(t) = 0.0000

Choosing the Lag Length in the ADF test

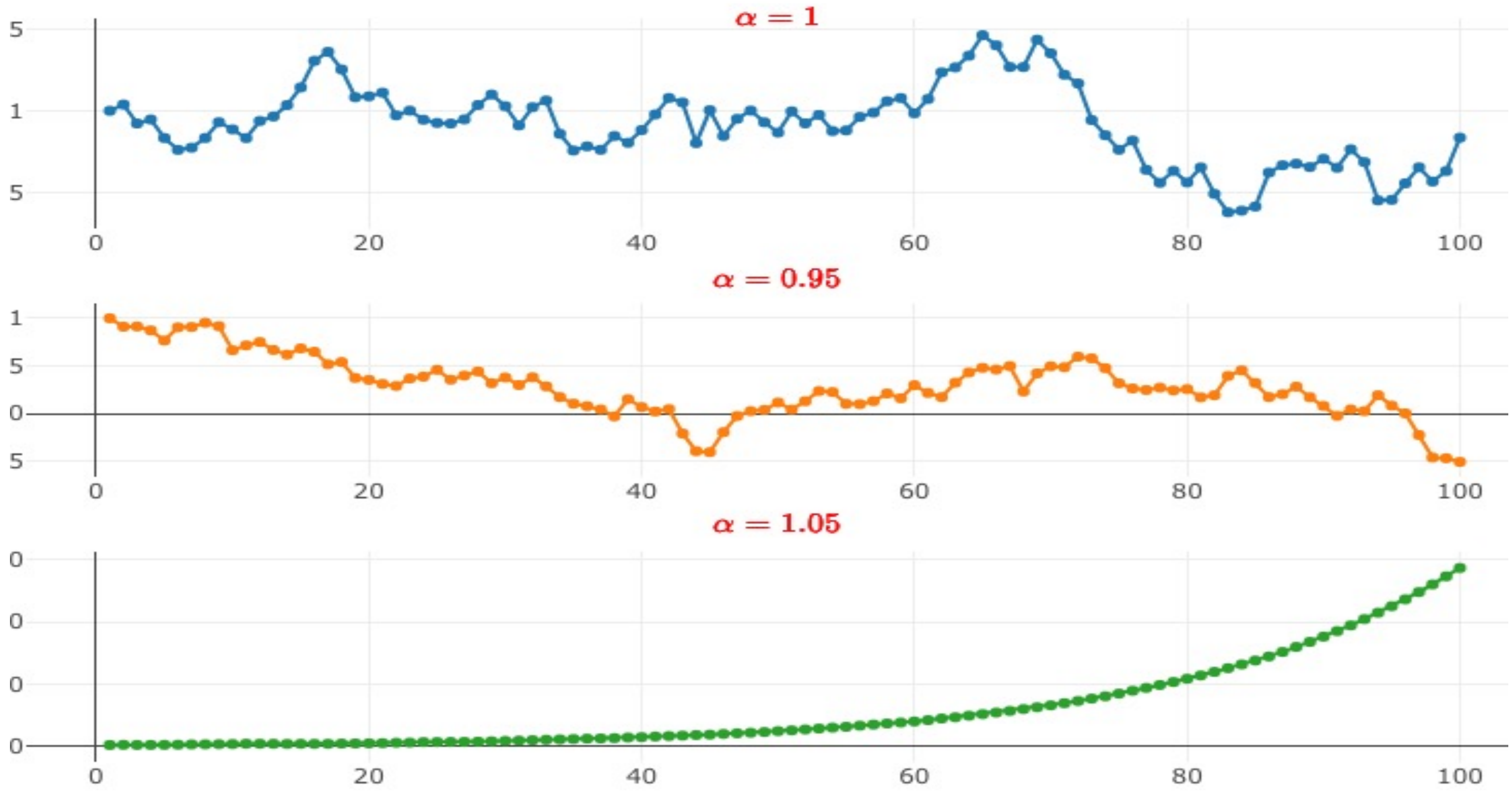
- Too few lags: the residuals will be autocorrelated and our results will be biased.

- Too many lags: reduces the power of the test to reject the null hypothesis.

- Campbell-Perron (1991) and Ng and Perron (1995): Sequential test method: start with a max k, if the t-stat on lag k-max is insignificant, re-estimate the regression using a lag length of k-1. Repeat the process until the lag is significantly different than zero. Ng and Perron (1995) compare the sequential test method with the AIC and found the sequential approach to be optimal (less size distortion and good power). Ng and Perron(2001) constructed a modified AIC (however, still the sequential approach perform better.
- Which k-max to choose? Schwert (1989, 2002) suggested the following formula:
- $kmax = int\left[12\left(\frac{T}{100}\right)^{1/4}\right]$
- Implemented on STATA: $kmax = int\left[12\left(\frac{T+1}{100}\right)^{1/4}\right]$
- Obs: Cheung and Lai (1995) found that the finite-sample critical values of the DF-GLS test depends on the number of lags (it is implemented on STATA).

Power of ADF and PP tests when the alternative is near-unit root process

- Both, ADF and PP tests have very low power against alternatives that are near-unit root process. It cannot distinguish the difference.
- The tests also lose power as deterministic terms are added to the test regressions.
- To improve the power of unit root tests against alternative that are near-unit root process, Elliot, Rothenberg and Stock (1996) propose a new procedure: DF-GLS. Also, the Ng-Perron sequential method should be used.



$\alpha = 1$
5
1
5
0
20
40
60
80
100
$\alpha = 0.95$
1
5
0
5
$\alpha = 1.05$
0
0
0
0

DF-GLS tests (command dfgls on STATA)

- Instead using OLS to detrend the data, Elliot, Rothenberg and Stock (ERS)(1992, 1996) proposed a GLS detrend procedure:

$$X_t = Z_t + v_t$$

$\Rightarrow Z_t = deterministic\ component$ = (1, t)

$$v_t = \alpha v_{t-1} + u_t$$

Let's

$$X_t^{\bar{\alpha}} = X_t - \bar{\alpha} X_{t-1}$$

$$Z_t^{\bar{\alpha}} = Z_t - \bar{\alpha} Z_{t-1}$$

$\bar{\alpha} = 1 + \frac{\bar{c}}{T}$ (local to unit alternative; with c<0)
$Z_0 = 0$

DF-GLS tests

$$X_t^{\bar{\alpha}} = [X_1, (1 - \bar{\alpha}L)X_2, \cdots, (1 - \bar{\alpha}L)X_T]$$

$$Z_t^{\bar{\alpha}} = [Z_1, (1 - \bar{\alpha}L)Z_2, \cdots, (1 - \bar{\alpha}L)Z_T]$$

- With$X_1^{\bar{\alpha}} = X_1 \;\; and \; Z_1^{\bar{\alpha}} = Z_1$

- $\bar{c} = -7$ *the case of only constant on the deterministic component.*
- $\bar{c} = -13.5 \;\; when\ the\ trend\ is\ included\ on\ the\ deterministic\ component$.

We then regress $X_t^{\bar{\alpha}}$ on $Z_t^{\bar{\alpha}}$, the estimated coefficient of $Z_t^{\bar{\alpha}}$ is the $\tilde{\beta}_{\text{GLS}}$ estimator. To detrend X_t:

$$\widetilde{X_t} = X_t - \widetilde{\beta_{GLS}} Z_t$$

After detrending X_t, we apply ADF test to $\widetilde{X_t}$.

DF-GLS tests

- The procedure removes the mean and linear trend for series that are not far from the non-stationary region (values of α close to 1). The unknown parameters of the trend function are efficiently estimated under the alternative model with $\bar{\alpha} = 1 + \frac{c}{T}$.

- Elliot and all show that the test will have a 50% power when the alternative is characterized by $\bar{\alpha} = 1 + \frac{c}{T}$; with $\bar{c}$ specified on the previous slide.
- ERS show that the DF-GLS test has the same asymptotic distribution of the ADF t-test when $\bar{c} = -0.7$. However, when trend is included in the deterministic component, the asymptotic distribution differ from the ADF t-test.

- DF-GLS test have higher power than ADF and PP unit root tests especially against local alternatives.
- Ng and Perron (2001) use the GLS detrending procedure to create modified version of the PP tests (called modified Phillips-Perron test). The modified tests have much smaller size distortions and more power.
- Kim-Perron() also uses the GLS detrending procedure to create unit-root tests that are robust againd the null and the alternative hypotheses.

dfgls UNRATE, maxlag(20)

DF-GLS for UNRATE Number of obs = 796

[lags]	DF-GLS tau Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
20	-3.586	-3.480	-2.825	-2.541
19	-3.430	-3.480	-2.827	-2.543
18	-3.222	-3.480	-2.830	-2.545
17	-3.183	-3.480	-2.832	-2.547
16	-3.186	-3.480	-2.834	-2.549
15	-3.122	-3.480	-2.836	-2.550
14	-3.001	-3.480	-2.838	-2.552
13	-2.921	-3.480	-2.840	-2.554

12	-2.966	-3.480	-2.842	-2.556
11	-3.370	-3.480	-2.844	-2.558
10	-3.291	-3.480	-2.846	-2.559
9	-3.705	-3.480	-2.848	-2.561
8	-3.736	-3.480	-2.850	-2.563
7	-3.764	-3.480	-2.851	-2.564
6	-3.984	-3.480	-2.853	-2.566
5	-3.991	-3.480	-2.855	-2.568
4	-3.558	-3.480	-2.857	-2.569
3	-3.240	-3.480	-2.858	-2.571
2	-2.706	-3.480	-2.860	-2.572
1	-2.133	-3.480	-2.862	-2.574

Opt Lag (Ng-Perron seq t) = 19 with RMSE .1891553

Min SIC = -3.242311 at lag 5 with RMSE .192756

Min MAIC = -3.264375 at lag 12 with RMSE .1900337

KPSS test

- It is normally used as a complementary test (to compensate for low power on previous tests):

- $\begin{cases} H_0\text{: } X_t \sim I(0) \\ H_1\text{: } X_t \sim I(1) \end{cases}$

- The KPSS test decomposes the time-series into the sum of a deterministic trend, a random walk component and stationary error:

$$X_t = \beta t + r_t + e_t$$

$$r_t = r_{t-1} + u_t$$

- r_0 plays the role of the intercept; $u_t \sim iid(0, \sigma_u^2)$
- e_t is a stationary error.

- If the variance of u_t is zero, then the value of r_t is always zero (assuming mean zero= r_0) $\Rightarrow r_1 = r_2 = r_3 \cdots = r_0$n this case, r_t is no longer a random walk and X_t is trend stationary model. Under the alternative, $\sigma^2 \neq 0$, and X_t is a random walk process.
- The KPSS test is a Lagranger Multiplier test (LM), under the null the statistic is: the ratio of two different estimates of the residual variance.

$$KPSS = \frac{\left(T^{-2} \sum_{t=1}^{T} \hat{S}_t^2\right)}{\hat{\sigma}_u^2}$$

Where: $\hat{S}_t = \sum_{i=1}^{t} \hat{u}_t$

$\hat{u}_t$ is the residual of a regression of X_t on the deterministic components.

And $\hat{\sigma}_u^2$ is the estimated of the long-run variance of the error using $\hat{u}_t$ (calculated using a particular weighing method (the Bartlett Kernel).

$$\hat{\sigma}_u^2 = \frac{1}{T}\sum_{i=1}^{T} \hat{u}_t^2 + \frac{2}{T}\sum_{s=1}^{l} w(s,l) \sum_{l=s+1}^{T} \hat{u}_t \hat{u}_{t-l}$$

Under the null: $\sigma_u^2 = 0$

Under the alternative: $\sigma_u^2 > 0$

kpss Y, maxlag(10) notrend auto

KPSS test for Y (Y is a random walk with drift)

Automatic bandwidth selection (maxlag) = 6
Autocovariances weighted by Bartlett kernel

Critical values for H0: Y is level stationary

10%: 0.347 5% : 0.463 2.5%: 0.574 1% : 0.739

Lag order	Test statistic
6	1.53

kpss X, maxlag(10) trend auto

KPSS test for X (X is trend-stationary)

Automatic bandwidth selection (maxlag) = 6

Autocovariances weighted by Bartlett kernel

Critical values for H0: X is trend stationary

10%: 0.119 5% : 0.146 2.5%: 0.176 1% : 0.216

Lag order	Test statistic
6	.0917

NELSON AND PLOSSER

```
dfuller lrgnp, trend reg lags(1)

Augmented Dickey-Fuller test for unit root        Number of obs   =       60

                          ---------- Interpolated Dickey-Fuller ---------
               Test        1% Critical       5% Critical      10% Critical
            Statistic         Value            Value            Value
------------------------------------------------------------------------------
 Z(t)           -2.994           -4.128           -3.490           -3.174
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.1338
------------------------------------------------------------------------------
    D.lrgnp |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
      lrgnp |
        L1. |  -.1753423   .0585665    -2.99   0.004     -.292665   -.0580196
        LD. |   .4188873   .1209448     3.46   0.001     .1766058    .6611688
     _trend |   .0056465   .0018615     3.03   0.004     .0019174    .0093757
      _cons |   .8134145   .2679886     3.04   0.004     .2765688     1.35026
------------------------------------------------------------------------------
```

```
 dfuller D.lrgnp, trend reg lags(1)
Augmented Dickey-Fuller test for unit root        Number of obs   =       59
                    ---------- Interpolated Dickey-Fuller ---------
              Test        1% Critical       5% Critical      10% Critical
            Statistic         Value            Value            Value
------------------------------------------------------------------------------
 Z(t)           -4.650          -4.130          -3.491          -3.175
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0008
------------------------------------------------------------------------------
   D2.lrgnp |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
  lrgnpLD. |    -.72593    .156109    -4.65   0.000    -1.038779   -.4130805
       LD2. |   .0890557   .1349329     0.66   0.512    -.1813558    .3594673
     _trend |   .0002638   .0004878     0.54   0.591    -.0007137    .0012413
      _cons |   .0133998    .017241     0.78   0.440    -.0211519    .0479515
------------------------------------------------------------------------------
```